GeoLabeller

Interface Control Document

Software version v1.2.0

Generated 2026-08-17

This document specifies the external interfaces of GeoLabeller: the project file it reads and writes, the raster products it exports, the imagery it accepts, and the coordinate conventions all of these share. It is a description of the implemented behaviour, transcribed from the source.

1 Scope and Conventions

GeoLabeller is a desktop application for viewing georeferenced and non-georeferenced raster imagery and recording point annotations against it. This document specifies the interfaces through which other software exchanges data with it.

1.1 Interfaces covered

Interface	Direction	Section
Project file (.geoproj / JSON)	Read and written	3
Sub-image GeoTIFF export	Written	4.1
Optimized GeoTIFF export	Written	4.2
Source imagery	Read	5

The HDF5 training-dataset export is specified separately and is not repeated here.

1.2 Notation

- Field types are JSON types: number, string, boolean, array, object.
- Pixel coordinates are (column, row) with the origin at the top-left pixel of the source image; column increases right, row increases down. Fractional values are meaningful and are not rounded on write.
- Geographic coordinates are WGS84 decimal degrees (EPSG:4326), longitude and latitude reported separately and never packed.
- Distances in metres are geodesic (Haversine on WGS84), not planar distances in a projected CRS.

2 Coordinate Systems

Four coordinate systems appear across the interfaces. Only the first two are externally visible; the rest are described because they explain values that would otherwise look surprising.

System	Units	Where it appears
WGS84 (EPSG:4326)	degrees	Label lon/lat, waypoints, image corner coordinates
Source pixel	pixels	Label pixel_x / pixel_y, relative to the unmodified source image
Web Mercator (EPSG:3857)	metres	Internal display projection; not written to any file
Scene	metres	Internal canvas space; x = easting, y = -northing

2.1 Pixel coordinates are relative to the source image

Labels record the pixel they were placed on in the ORIGINAL image, at full resolution - not in the reprojected image shown on screen, and not at whatever pyramid level happened to be displayed. A label therefore addresses the same ground whatever the zoom was when it was placed, and a consumer can index the source raster directly.

2.2 Non-georeferenced imagery: the pixel zone

Imagery without a CRS cannot be placed geographically, so it is laid out in a reserved region of the internal scene beyond the valid extent of Web Mercator. This is an implementation detail with one external consequence: labels on such images carry pixel coordinates but no meaningful lon/lat, and those fields are written as 0.0.

Constant	Value	Meaning
PIXEL_ZONE_ORIGIN_X	25 000 000	Scene X at which the non-georeferenced region begins
PIXEL_ZONE_SCALE	50	Scene units per source pixel
PIXEL_ZONE_GROUP_GAP	5 000	Scene units between the columns of adjacent groups

2.3 Display resolution and memory

Imagery is displayed from pyramid overviews. The level chosen is the coarsest whose resolution is still no coarser than one screen pixel, subject to a cap on how much may be held at once. Images beyond that cap keep a coarse whole-image backdrop and read the detail for the visible area one tile at a time, so full resolution remains available at any zoom regardless of image size.

Constant	Value	Meaning
MAX_LEVEL_PIXELS	150 000 000	Largest whole-image level held in memory
BACKDROP_MAX_PIXELS	4 000 000	Cap for the backdrop behind windowed detail tiles
TILE_SIZE	512	Edge of a display tile, in destination pixels

3 Project File

A project is a single UTF-8 JSON document holding the class list, every image referenced, the labels on them, and any waypoints. Images are referenced by absolute path; their pixels are never copied into the project.

3.1 Top level

Field	Type	Description
version	string	Format version of this document, currently "3.3". Governs how label coordinates are interpreted - see 3.4.
classes	array of string	Ordered class names. A label's class must appear here.
images	array of object	One entry per referenced image; see 3.2.
waypoints	array of object	Named geographic bookmarks; see 3.5. Absent before version 3.3.
_next_id	number	Next label id to allocate. Labels ids are unique per project.
_next_waypoint_id	number	Next waypoint id to allocate.

3.2 Image object

Field	Type	Description
path	string	Absolute path to the image file.
name	string	File name without extension.
group	string	Group hierarchy, '/' separated; empty when ungrouped.
labels	array of object	Labels on this image; see 3.3.
original_width	number	Source image width in pixels. Required to interpret label coordinates - see 3.4.
original_height	number	Source image height in pixels.
reader	object	Extension to reader name, e.g. {"tif": "default"}.
affine_coeffs	array of number	Six coefficients of the source affine transform. Omitted when unknown.
crs_epsg	number	EPSG code of the source CRS. Omitted for non-georeferenced imagery.
corners_wgs84	object	Convenience only: top_left / top_right / bottom_right / bottom_left, each {lat, lon}. Written when the transform and CRS are known; derived, never read back.
geodesic_width_m	number	Convenience only: ground width along the top edge, metres.
geodesic_height_m	number	Convenience only: ground height along the left edge, metres.

3.3 Label object

Field	Type	Description
id	number	Sequential id, unique within the project.
unique_id	string	UUID v4, unique to this label.
class_name	string	Must appear in the top-level classes.
pixel_x	number	Column in the source image. Stored as a FRACTION of original_width from version 2.1 - see 3.4.
pixel_y	number	Row, stored the same way.
lon	number	WGS84 longitude, or 0.0 for non-georeferenced imagery.
lat	number	WGS84 latitude, or 0.0.
object_id	string	UUID v4 shared by every label of the same real-world object. Labels are linked by sharing this value.
length_m	number	Measured object length in metres. Omitted when unmeasured.

Field	Type	Description
width_m	number	Measured width in metres. Omitted likewise.
geodesic_distance	number	Convenience only: metres from the image's left edge at the label's row. Derived, never read back.

3.4 Label coordinate storage (important)

From version 2.1, pixel_x and pixel_y are stored as FRACTIONS of the image dimensions, not as absolute pixels. A consumer must multiply by the image size to recover pixel coordinates:

```
pixel_x_absolute = pixel_x * original_width
pixel_y_absolute = pixel_y * original_height
```

The conversion applies only when the document version is 2.1 or later AND original_width and original_height are both greater than zero. If either is zero the values are absolute pixels and must be used unchanged. Both the writer and the reader apply this rule, so a file is self-consistent; it matters when reading a project with anything other than GeoLabeller itself.

3.5 Waypoint object

Waypoints are named geographic bookmarks belonging to the project rather than to any image. They are navigation aids and are never exported as training data.

Field	Type	Description
id	number	Sequential id, unique within the project.
name	string	Display name, e.g. "WP 1".
lat	number	WGS84 latitude in degrees.
lon	number	WGS84 longitude in degrees.

3.6 Version history

Version	Change
1.0	Label-centric layout: a flat labels array, each entry naming its image. Still readable.
2.0	Image-centric layout: images each own their labels.
2.1	Label pixel coordinates became fractions of the image dimensions (see 3.4).
3.2	Adds derived convenience fields: corner coordinates and geodesic extents.
3.3	Adds the top-level waypoints array and _next_waypoint_id. Files written before this simply have none.

Readers should compare the version as a string and treat anything at or above a threshold as supporting that feature. Older documents load unchanged; there is no migration step.

3.7 Example

A project with one image, one label and one waypoint. Note the fractional pixel coordinates: label 7 sits at absolute pixel (2048, 1024), being 0.5 x 4096 and 0.25 x 4096.

```
{
  "version": "3.3",
  "classes": ["vehicle", "building"],
  "images": [
    {
      "path": "D:/imagery/tile_014.tif",
      "name": "tile_014", "group": "survey/north",
      "original_width": 4096, "original_height": 4096,
      "reader": {"tif": "default"}, "crs_epsg": 32618,
      "labels": [
        {"id": 7, "class_name": "vehicle",
         "pixel_x": 0.5, "pixel_y": 0.25,
         "lon": -73.9832, "lat": 40.7536,
         "unique_id": "3b1e...", "object_id": "8f2c...",
         "length_m": 4.6, "width_m": 1.8}
      ]
    }
  ],
  "waypoints": [{"id": 1, "name": "WP 1",
                "lat": 40.7536, "lon": -73.9832}],
  "_next_id": 8, "_next_waypoint_id": 2
}
```

4 Raster Exports

4.1 Sub-image export

Writes one GeoTIFF per label, cropped around it. The requested size is given in ground metres and converted to pixels using the true resolution at the label, so the crop covers the requested ground area whatever the CRS and whatever the pixel aspect.

Property	Behaviour
Output path	<output>/<class_name>/<object_id>_<label_id padded to 6>.tif
Window	Centred on the label's pixel, shifted (never cropped) to stay inside the image, so edge labels still yield a full-size crop.
Pixels	Copied unmodified: all source bands, original data type, no rescaling.
Georeferencing	Source CRS preserved; transform adjusted to the window.
Other metadata	nodata and per-band colour interpretation preserved.
Compression	LZW.
Skipped	Labels on imagery with no CRS, labels outside the image, and windows that would be under half the requested size.

4.2 Optimized GeoTIFF export

Rewrites imagery as tiled, pyramided GeoTIFFs for fast display. Pixel values, band count, data type, CRS and transform are all preserved - only the storage layout changes.

Property	Behaviour
Output path	Mirrors the group hierarchy under the output directory; keeps the source stem with a .tif extension.
Layout	Tiled, 512 x 512 blocks by default.
Overviews	Built to the requested decimation factors. Average resampling by default; paletted rasters always use nearest, so palette indices are never blended into invalid values.
Compression	DEFLATE by default; may be set to another codec, none, or KEEP to retain the source's. A predictor is chosen to suit the data type.
Large files	BIGTIFF written when the result may exceed 4 GB.
Existing files	Skipped unless overwrite is requested. Output is written to a temporary file and moved into place, so an interrupted export leaves no partial .tif.

5 Source Imagery

Imagery is read through GDAL, so any format GDAL can open may be loaded; file dialogs filter to .tif and .tiff by default, and directory import scans for those extensions.

5.1 Georeferenced and non-georeferenced imagery

An image with a CRS is reprojected for display and its labels carry both pixel and WGS84 coordinates. An image without one is shown at its own pixel scale in the region described in 2.2, and its labels carry pixel coordinates only, with lon and lat written as 0.0.

5.2 Recommendations

- Supply tiled GeoTIFFs with overviews. Without a pyramid the whole image must be decoded before anything can be shown; with one, a coarse level appears immediately and sharpens. Section 4.2 produces suitable files.
- 8-bit imagery is displayed exactly as stored. Other data types are linearly stretched for display between the 2nd and 98th percentile of a sampled read; the underlying values are never modified.
- Very large mosaics are supported: detail for the visible area is read a tile at a time, so display cost follows the viewport rather than the size of the image.